

Iván Cruces
Málaga (Spain)
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Professional Summary

Software Engineer with 18+ years of experience building distributed object storage systems and full-stack applications. Experienced in distributed systems, fault-tolerant architectures, data security, search, and production troubleshooting. Strong background in Java, storage platforms, and backend API development.

Employment History

DataCore (*Málaga, Spain*)

Staff Software Engineer

Feb 2023-Oct 2026

- Built a **multi-tenant REST API for file management** in a distributed storage platform, with OAuth2/Keycloak authentication, configurable S3 storage backends, and operations including multipart uploads. Integrated with **search and management APIs** to provide file discovery and enforce user permissions and access controls.
- Developed a **multi-tenant search API** backed by Elasticsearch.
- Integrated **AI-generated metadata** into the search platform via RabbitMQ, enabling search across **video transcriptions, facial recognition and object recognition** results.
- Led the **GitLab-to-GitHub migration**, developing API-based migration tooling, reimplementing CI/CD pipelines as **GitHub Actions**, and configuring self-hosted runners.
- Defined the architecture for an **Angular application** and provided ongoing technical guidance through **code reviews and design discussions**.

Object Matrix Ltd. (*Cardiff, UK – Málaga, Spain*)

Senior Software Engineer

April 2012-Feb 2023

- Developed a distributed search engine using Apache Lucene, coordinating searches across indexes distributed across multiple storage nodes, including result aggregation and crash recovery using a transaction log.
- Designed and implemented cross-cluster data replication for **disaster recovery**. Enhanced **data integrity** and **access control**, and implemented **compliance** and **audit features**.
- Led the modernization of legacy components through full-stack development with Angular and Spring Boot, building REST APIs and web applications for file browsing and cluster administration.
- Investigated and resolved complex customer issues through direct troubleshooting of customer systems and production environments.

Artificial Intelligence research group, University of Málaga. Spain

Project researcher

2008-2012

- Developed an **adaptive e-learning system** integrated with Moodle, recommending learning activities based on students' inferred knowledge models.
- Developed **Java EE web services**, relational database systems, and XML-based applications.

Asamiol. Cádiz, Spain

Computer teacher

2007-2008

- Taught fundamental computer skills and developed tailored curricula for learners with diverse needs.

Education

**Master's Degree in Software Engineering
and Artificial Intelligence**

2009-2011

University of Malaga. Spain

Degree in Computer Engineering (Master's level, EQF 7)

2001-2008

University of Malaga. Spain

Skills Summary

Programming languages: Java, Bash, JavaScript, Python

Distributed Systems & Storage: S3, Riak, Hazelcast, Elasticsearch, Lucene

APIs, Frameworks & Web: Spring, Angular, Django, REST APIs, HTML, CSS

Messaging & Security: RabbitMQ, Keycloak, OAuth2

Tools & DevOps: Git, Docker, Maven, Gradle, GitLab CI, GitHub Actions

Language Skills

Spanish: Native.

English: Professional working proficiency

Publications

I. Cruces, M. Trella, R. Conejo and J. Gálvez: “**Student Modeling Services for Hybrid Web Applications**”, in Proceedings of International Workshop on Architectures and Building Blocks of Web-Based User-Adaptive Systems (UMAP 2010), 2010.